

UMA012: NUMERICAL AND STATISTICAL COMPUTATIONS

L	T	P	Cr
3	0	2	4.0

Course Objective: The main objective of this course is to understand and implement various numerical and statistical methods to solve civil engineering problems.

Basic of Errors: Floating-point representation, rounding and chopping errors.

Non-Linear Equations: Bisection, Newton - method for simple and multiple roots and order of convergence.

Linear Systems and Eigen-Values: Gauss elimination method using partial pivoting, Gauss-Seidel method, Rayleigh's power method for eigen-values and eigen-vectors of matrix.

Interpolation and Approximations: Lagrange (with error analysis), Newton's divided difference and forward difference interpolation formulas.

Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis.

Differential Equations: Solution of initial value problems using Euler's, Modified Euler's and Runge - Kutta methods (fourth-order), Solution of boundary value problems using finite difference method.

Curve Fitting and Regression: Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas, Bivariate distribution, correlation coefficients, regression lines, formula for regression coefficients and rank correlation (only two variables).

Probability Distribution: Mathematical expectations, Definition of probability distribution (Probability Mass Function and Probability Density Function), Poisson, Binomial, Exponential, Uniform and Normal distributions.

Laboratory Work: Lab experiments will be set in consonance with materials covered in the theory. Implementation of numerical and statistical techniques using MATLAB to solve following civil engineering problems:

1. Non Linear Equation: A simply supported beam problem to find the point (s) of zero shear and zero moments.

2. Linear System and Eigen Values: Column problem subjected to axial load and bi-axial moments and to find the principal normal stresses and the direction cosine of the normal principal stresses. Calculate the forces in each members of a truss subjected to given loading.

3. Interpolation and Approximation: Find the Water levels of reservoir at intermediate points.

4. Numerical Integration: Find the capacity of a reservoir.

5. Differential Equation: Finding the solution of a single degree of freedom (SDOF) subjected to free and forced vibration.

6. Population growth problem.

Course Outcomes: Upon successful completion of the course, the students will be able to

1. obtain numerical solution of nonlinear equations using bisection and Newton-Raphson methods.
2. solve system of linear equations numerically using direct and iterative methods.
3. construct interpolating polynomials with practical exposure and analyze the correlated data using the
4. linear regression models.
5. solve integration and initial value problems numerically. .
6. solve civil engineering problems using various probability distributions.

Text Books:

1. E. Ward Cheney and David R. Kincaid, Numerical Mathematics and Computing, Cengage, 2013.
2. Walpole, Ronald E., Myers, Raymond H. Myers, and Sharon L. Myers, Probability and Statistics for Engineers and Scientists, 9th edition Pearson Education, 2012

Reference Books:

1. Burden L. R., Faires D. J. and Burden A.M., Numerical Analysis, Brooks Cole, 10th edition, 2015.
2. Steven C. Chapra and Raymond P. Canale, Numerical Methods for Engineers, McGraw-Hill Higher Education; 7th edition, 2016.
3. M.K. Jain, S.R.K. Iyengar, and R.K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, 2003.
4. Richards A. Johnson, Probability and Statistics for Engineers, 9th Edition, PHI Learning, 2018.

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessment and Quizzes, Project (Including report, presentation etc.)	40